

Text Handoff Lauanders Uncertainty; A Mapped KV Cache Keeps More of It

Anonymous Authors¹

Abstract

When one language model hands work to another through text, the receiver gets the sender’s answer but not how sure the sender was: a text handoff is a point estimate, and the receiver treats it as a fact. We ask whether handing over the sender’s key–value (KV) cache instead keeps that uncertainty. Between two frozen models of different size, K2-Horizon-3.7B and K2-Horizon-7B, we train a per-head map from the small model’s cache to the large model’s, and test on SQuAD passages that are clean, contradicted, or have their answer removed. The receiver’s confidence through the mapped cache tracks the *sender’s* semantic entropy about as well as when the receiver reads the text itself (AUROC .585 / .605 / .624 vs .597 / .615 / .671), and better than a text handoff in which the sender adds its answer and a confidence word (.552 / .542 / .592); when the answer is removed, the cache path loses 54% of the confidence the text path loses where the verbal handoff loses 30%, and it answers better on clean and contradicted passages (F1 59.8 / 49.0 vs 55.0 / 37.7). The precondition is that the cache carries content at all, which we establish with controls: a closed-form linear map recovers almost nothing on a binding test (22%, chance 12.5%), a gated residual trained on the receiver’s continuation loss reaches 63–73%, the receiver follows a cache from the wrong episode 64% of the time, a projector trained with a deranged sender answers at 23%, and on SQuAD the mapped cache reaches 53.0 F1 against 54.7 for reading the passage. The same recipe crosses model families (Qwen3-4B → 7B: 76.7%) and fails for a same-family sender with a different vocabulary and head dimension (K2-0.9B: 17%). A first attempt to train the confi-

dence loss away with a KL term did not work; we say why. We release the maps and the tests.

1. Introduction

Edge–cloud agent stacks run a small model on device and escalate to a large one when the small model is unsure. Escalation today means writing a message: the small model’s answer, perhaps with a confidence word, and the large model reads it and continues. The message is a point estimate. Two senders with the same best guess and very different spread send the same words, so whatever the small model knew about its own uncertainty is discarded at the boundary, and the receiver treats a confident-looking guess as a fact—the “confidence laundering” that [Shi et al. \(2026\)](#) describe. If the large model could instead continue from the small model’s internal state, the handoff would carry what the small model had computed, including its spread. This paper measures how much of the sender’s uncertainty survives each kind of handoff, and finds that a mapped KV cache keeps more of it than text with a confidence word.

Before uncertainty can cross, content must. Given two frozen models of different size that share a tokenizer and KV geometry (K2-Horizon-3.7B and 7B: 36 layers, 8 KV heads, head dimension 128), can a learned map from the small model’s cache to the large model’s cache carry a *specific fact* that only the small model saw? We call this binding transfer, and we test it with a control for the standard objection that “latent gains” are extra compute rather than communication. The content results are the foundation; the uncertainty result is what the foundation is for.

Contributions. (1) A sender-uncertainty test on SQuAD with clean, contradicted and answer-removed passages, and the finding that a mapped-cache handoff tracks the sender’s semantic entropy better than a text handoff with a confidence word and loses less confidence when the answer is gone (Section 4), with a short argument for why a text handoff cannot do better (the theory box). (2) A binding-transfer test (Noma) with six arms that isolates content from shape and compute, and a two-stage projector—closed-form ridge over correlation-selected source layers plus a gated residual trained on the receiver’s continuation loss—that reaches

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

63–73% binding transfer where the ridge alone reaches 22% (Section 3). (3) Controls that bound what the receiver actually gets from the cache: a deranged-episode cache, a deranged-sender projector, layer-band injection, and the demonstration that continuation retention, the usual headline, cannot tell a channel from a prompt (Section 5). (4) Cross-family, cross-context and message-cost results that say where the channel works and what it costs.

2. Setup

Models. Sender K2-Horizon-3.7B, receiver K2-Horizon-7B, both frozen, bf16, RoPE $\theta = 10^7$. We capture the sender’s pre-RoPE K and raw V at every layer, map them, apply RoPE at the receiver, and insert the result as the receiver’s cache for the prefix positions. The receiver then reads only the question.

Projector. Stage 1: for each receiver layer ℓ and KV head h , a ridge regression from the concatenated $[K; V]$ of the sender (256-d per head) to the receiver’s $[K; V]$; source layers are the three sender layers whose K/V correlate best with receiver layer ℓ (“top-3”), which beat the aligned-layer map in a first pass. Stage 2: a gated SwiGLU residual (bottleneck 128, 151 M parameters) on top of the ridge output, trained with AdamW for 1,500 steps, effective batch 16 sequences of 1,024 FineWeb-Edu tokens, prefix 512, on the receiver’s cross-entropy over the continuation. Nothing in training mentions the tests.

Noma binding test. Each episode assigns one of eight single-token values to eight nonce names (“Noma is amber. Vela is indigo. ...”). Only the sender reads the facts. The receiver is asked “What colour is Vela?” and we score the eight values. Arms: *none* (question only), *text* (receiver reads the facts itself), *self* (receiver’s own cache re-injected; equals *text* to 10^{-4}), *raw* (sender cache injected without a map), *project* (mapped cache), *derange* (mapped cache from another episode with the same layout but a different value for the queried name). Content transfer = project – derange. For *derange* we also report the follow rate: how often the answer is the *other* episode’s value.

Passages. SQuAD v1.1 validation: the sender reads the passage (≤ 512 tokens), the receiver reads only the question and answers greedily (≤ 8 tokens, stopped at newline); token-F1 against the gold spans. The uncertainty protocol built on it is given in Section 4.

3. Content Transfer: The Precondition

Binding transfer. Table 1 and Figure 1 give the main content result. On the original 6-name variant ($n = 300$) three training seeds give 71.3 / 73.0 / 75.3% (mean 73.2, sd 2.0). Two ablations on the same variant (Appendix A): freez-

Table 1. Binding transfer, 8 names $\times$ 8 values, $n = 1000$ episodes, chance 12.5%. All entries are percentages. “follow” is the rate at which the receiver, handed the deranged cache, answers with the other episode’s value; “content” is project – derange.

projector	none	text	raw	project	derange	follow	content
ridge top-3 (closed form)	10.6	100	14	21.7	11.4	18.2	10.3
+ residual, 1500 steps	10.6	100	14	63.0	4.6	63.7	58.4
residual, deranged sender	10.6	100	14	23.4	11.3	22.7	12.1

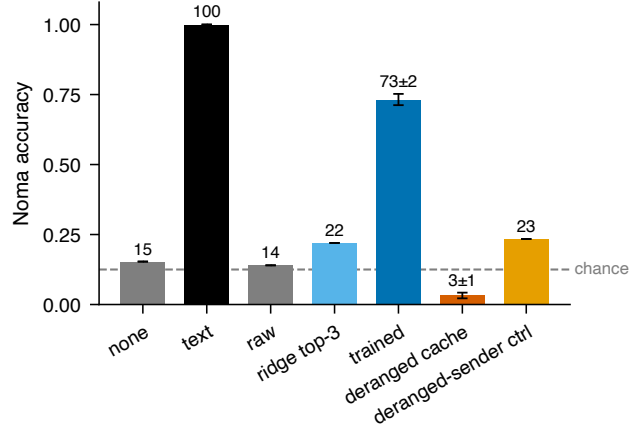


Figure 1. Noma accuracy by arm on the 6-name variant ($n = 300$). “trained” and “deranged cache” are the mean over three training seeds with error bars showing one standard deviation; the deranged-sender control is the *project* arm of the projector trained with a deranged sender ($n = 1000$, 8-name variant). Dashed line: chance (12.5%).

ing the ridge and training only the residual gives 59.0%, so letting the linear map move during training is worth ~ 14 points; replacing the correlation-selected top-3 source layers with the aligned-layer map gives 32.7%, so which sender layers feed each receiver layer is the single largest design choice. Matching shape alone (*raw*) does nothing. The follow rate is the decisive number: when the receiver is handed the wrong episode’s cache it answers with the wrong episode’s value 64% of the time, and its accuracy falls below chance. The receiver believes the cache. With the full 8-name pool and 500 episodes, the seed-2 projector gives 65.8% and 67.6% on two episode seeds (ridge: 20.4%, 20.6%). On 256 fresh held-out sequences deduplicated against training text, continuation retention is 1.19 for the trained projector and -0.38 for the ridge. The same protocol with cities and animals gives 61.8% and 67.2% (Appendix B), and padding the facts with up to 512 filler tokens does not reduce accuracy (61.0 / 57.0 / 62.4 / 68.8% at 64 / 128 / 256 / 512; Appendix C).

Where the binding lives. Table 2 (plotted in Appendix C, Figure 7b): the first half of the network carries nothing usable on its own; the second half alone carries most of it, and no six-layer band suffices. The binding is read out from a distributed set of later layers.

Table 2. Where the binding lives. This episode’s mapped cache is injected only into a band of receiver layers, the deranged episode’s elsewhere. $n = 500$, 6-name variant; accuracy in percent.

layers	0–5	0–11	0–17	12–23	24–35	30–35	18–35	all
accuracy	7.2	8.0	10.4	28.6	30.8	7.6	51.8	63.0

Table 3. Real passages: SQuAD v1.1 validation, $n = 300$ (600 for the trained projector). Token-F1 against the gold spans; mean gold log-probability per token.

receiver sees	none	text	raw	ridge	trained	deranged-sender ctrl	trained, derange arm
F1	16.6	54.7	0.6	14.3	53.0 (51.8)	15.1	13.8
gold log-prob / token	−3.57	−0.90	−11.2	−3.88	−1.74	−3.77	−4.18

Real passages. Table 3 and Figure 2 apply the same projectors, never trained on questions, to SQuAD. On F1 the trained projector recovers 96% of the gap between question-only and reading the passage (53.0 at $n = 300$; 51.8 at $n = 600$); on gold log-probability it recovers 69%. The wrong passage’s cache (derange), the ridge map, and the content-free projector all sit at or below question-only. The channel carries the passage, not a bias toward answering.

Across pairs and families. Table 4 and Figure 3. The channel is not one-way: with the roles swapped (7B sender, 3.7B receiver) the ridge map gives 29.0% and the trained projector 63.3%, with the deranged cache at 5.7% and a 63% follow rate—the same pattern at slightly lower accuracy. With Qwen3-4B as the sender (36 layers, 8 KV heads, head dimension 128—matched KV geometry—but a different vocabulary and pre-training), receiver positions are aligned to sender tokens by character offsets and the same recipe is run: top-3 ridge 18.0% (raw injection 15.7%, deranged 13.0%), trained projector 76.7% (deranged 1.3%, follow rate 80%), SQuAD F1 48.0 against 54.7 for the receiver reading the text. Cross-family transfer is as strong as within-family on the binding test and within 7 F1 of it on passages, and the linear map alone does nothing across families. The recipe fails for K2-Horizon-0.9B, a same-family sender with a different vocabulary and head dimension 64: the trained projector reaches 17.0% (deranged 14.0%, follow 11.3%; content transfer 3 points) and 24.2 F1 on SQuAD. Matched head dimension, not shared pre-training, is what the per-head map needs.

Different contexts. Table 5: HotpotQA bridge questions, sender and receiver each holding one of the two supporting paragraphs (the LCF-X setting of Rossi et al., 2026). The receiver with its own paragraph and the question reaches F1 30.1; with both paragraphs as text 45.8; with its own paragraph plus the sender’s mapped cache 37.9—half of the gap—while the deranged cache gives 29.9 (no help), the ridge map 26.3 and a zero cache 20.9. The channel carries the *other* paragraph, and only the trained map does.

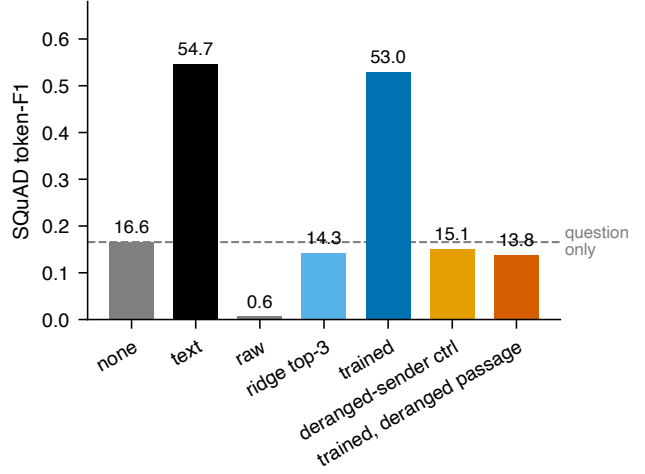


Figure 2. SQuAD token-F1 by arm ($n = 300$). Dashed line: question only. The trained projector sits within two points of the receiver reading the passage itself; the ridge map, the deranged-sender control, and the trained projector given the wrong passage’s cache all sit at or below question-only.

Table 4. Four sender–receiver pairs, same recipe. Noma 6-name variant, $n = 300$, chance 12.5%; percentages. “proj.” is the *project* arm, “der.” the *derange* arm, “follow” the follow rate. SQuAD F1 is for the trained projector ($n = 300$; text ceiling 54.7). K2-Horizon unless named. The raw arm is undefined for K2-0.9B (head dimension 64).

sender → receiver	raw	ridge top-3			+ trained residual			F1
		proj.	der.	follow	proj.	der.	follow	
3.7B → 7B	14.0	22.0	10.7	30.3	71.3	4.3	71.7	53.0
7B → 3.7B	11.7	29.0	12.3	31.3	63.3	5.7	63.0	—
Qwen3-4B → 7B	15.7	18.0	13.0	17.0	76.7	1.3	80.3	48.0
0.9B → 7B	n/a	13.3	13.0	10.7	17.0	14.0	11.3	24.2

What the message costs. The full mapped cache is 36 layers $\times$ 8 heads $\times$ 2 $\times$ 128 $\times$ bf16 = 144 KB per token, 75 MB for a 512-token prefix, against $\sim$ 1 KB of text; the projector has 151 M parameters. Table 6 and Figure 4: two things shrink it for free and two do not. int8 and int4 fake-quantisation of the mapped cache keep binding accuracy (76.0 / 75.0 vs 75.3%) and SQuAD F1 (52.1 / 51.6 vs 53.0)—bytes $\div$ 4. A projector with bottleneck 32 has 70 M parameters and keeps SQuAD F1 (53.7) while losing 9 points on the binding test (64.3%); bottleneck 16 (62 M) gives 61.3% and 51.6 F1. Dropping layers or heads with zeros in their place collapses transfer (layers 18–35 only: 24.3%; 4 heads: 25.7%), unlike the layer-band experiment where the other layers carried a *plausible* cache; an all-zero layer makes the receiver attend uniformly to nothing and drowns the question. Filling untransmitted layers with a one-vector summary (the position-mean of that layer’s mapped cache) recovers only part of it (layers 18–35: 33.7%; layers 12–35: 35.0%; SQuAD F1 21–29), so with this projector the message cannot be cut by layers: every receiver layer wants its own cache. Low-rank truncation of each (posi-

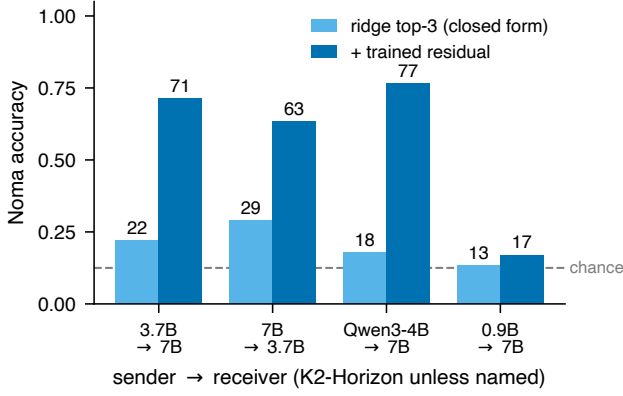


Figure 3. Noma accuracy (*project* arm) for the closed-form ridge map and the trained projector on four sender–receiver pairs ($n = 300$, 6-name variant). Dashed line: chance. The trained map crosses families (Qwen3-4B) but not a vocabulary and head-dimension change within the family (K2-0.9B).

Table 5. Different contexts: HotpotQA bridge questions, $n = 400$. Sender and receiver each hold one of the two supporting paragraphs; the receiver always reads its own paragraph and the question. Token-F1.

receiver additionally gets	F1
nothing (own paragraph + question)	30.1
the sender’s paragraph as text	45.8
the sender’s paragraph as a mapped cache	37.9
a deranged mapped cache	29.9
the sender’s paragraph through the ridge map	26.3
a zero cache	20.9

tions $\times$ head-dim) matrix keeps the binding test (rank 32 / 16: 75.0%) but not passages (F1 45.3 / 29.6). Latency on one A100: the receiver re-reading 512 tokens takes 48 ms, the projector 24 ms ($2.0\times$; the sender’s own read, 45 ms, is already paid). The message is therefore cheap in time and expensive in bytes; the honest framing is that the channel is worth it when the sender’s state is the point—which the next section tests.

4. Does the Cache Carry How Sure the Sender Was?

Why a text handoff launders uncertainty, and what a state handoff can keep. Sender S reads context c and holds a belief $p_S(y | c)$ over answers; it hands off through a message $m = f(c, S)$, and the receiver produces $p_R(y | m, q)$. With $m = \arg \max_y p_S(y | c)$, m takes at most $|Y|$ values, so the information m carries about p_S is bounded by $\log |Y|$ nats however peaked or flat p_S is: two senders with the same mode and entropies 0.1 and 2.0 send the same message. A verbalised confidence word adds at most $\log k$ nats ($k = 3$ levels here) and relies on the sender’s verbal calibration, which is poor for base models (Tian et al., 2023; Yona et al., 2024). This is the mechanism behind confidence laundering (Shi et al., 2026).

Table 6. What the message costs. Message size is bytes per prefix token of the mapped cache (512-token SQuAD passages; “KB” = 1024 bytes). Noma: 6-name variant, $n = 300$, seed-2 projector (full cache 75.3%). Rows with a smaller projector are separate 1,000-step training runs. The full sweep is in Appendix D.

message	KB / token	Noma	SQuAD F1
full mapped cache, bf16	144	75.3	53.0
int8 fake-quantised	72	76.0	52.1
int4 fake-quantised	36	75.0	51.6
low-rank, rank 32	78.7	75.0	45.3
low-rank, rank 16	39.3	75.0	29.6
layers 18–35 only, zeros elsewhere	72	24.3	8.3
4 of 8 heads, zeros elsewhere	72	25.7	8.2
layers 18–35, mean-vector fill	72.7	33.7	21.4
layers 12–35, mean-vector fill	96.4	35.0	28.8
projector bottleneck 32 (70 M params)	144	64.3	53.7
projector bottleneck 16 (62 M params)	144	61.3	51.6

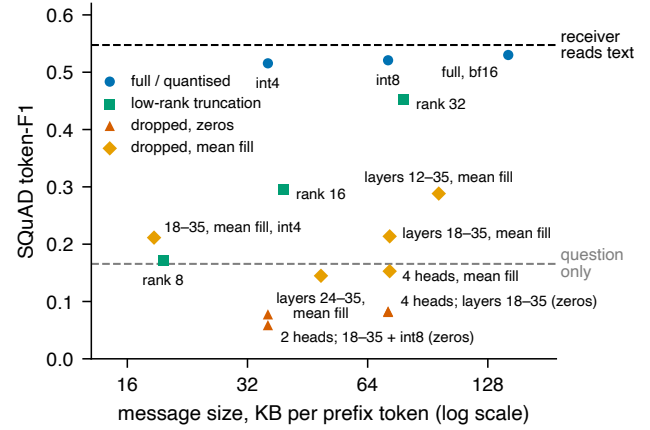


Figure 4. Message size against SQuAD F1 for every compression of the mapped cache we tried ($n = 300$). Quantisation is free; low-rank truncation and dropping layers or heads are not. The two “zeros” points at 72 and 36 KB each hide two coincident specs.

With m the KV cache of S at every layer and position, m is a sufficient statistic for S ’s next-token distribution, so $I(m; p_S)$ is not bounded by $\log |Y|$. What the receiver recovers is limited by two losses: *map loss*—the map g is fit on continuation likelihood, and a learned map trained on point targets contracts predictive variance (Cui & Pei, 2026); and *read loss*— R reads the mapped cache with its own heads, from its later layers (Table 2). Predictions: (1) the verbal handoff cannot beat its $\log k$ bound, so its AUROC against the sender’s semantic entropy sits near the tercile-label ceiling; (2) the cache handoff can exceed it and approach the text-reading receiver’s, its shortfall being map loss; (3) neither can exceed the ceiling set by how far R ’s first-token entropy correlates with S ’s semantic entropy at all, which is why absolute AUROCs are ~ 0.6 and the comparison must be relative.

Protocol. Following the knowledge-conflict and semantic-entropy protocols (Longpre et al., 2021; Xie et al., 2024; Farquhar et al., 2024), each SQuAD item gets three passages: *clean*, *contradicted* (one fluent sentence asserting a counter-answer, written by the 375B judge and kept only if DeBERTa-MNLI says it entails the counter-answer and not the gold: 587 of 941 survive), and *removed* (the gold

sentence deleted). We keep the 377 items the receiver answers correctly from clean text. The sender’s uncertainty is its semantic entropy over 10 sampled answers (mean 1.10 / 1.23 / 1.96 across the three variants, so the sender does notice). The receiver’s uncertainty is its entropy at the first answer token. The text handoff with a confidence word (*verbal*) is the text channel doing its best to carry uncertainty: the sender writes its greedy answer and “confidence: high / medium / low” (terciles of its own semantic entropy), and the receiver reads only that. The drop ratio is the receiver’s loss of $P(\text{gold})$ from clean to removed, relative to the text path’s loss.

Result. Table 7 and Figure 5. The mapped cache beats the verbal handoff on every measure: it tracks the sender’s uncertainty better (AUROC +.03 to +.06, on all three variants), it loses 54% of the confidence the text path loses on unanswerable passages where the verbal handoff loses 30%, and it answers better on clean and contradicted passages (F1 59.8 / 49.0 vs 55.0 / 37.7). A one-word confidence is a log 3 nat channel; the cache is not so limited, and the receiver uses the difference.

Three readings. (i) The receiver’s confidence through the cache tracks the *sender’s* semantic entropy about as well as it does when the receiver reads the text itself (AUROC .585 vs .597 on clean; every content-free control sits at .46–.54), and the absolute level is bounded by the text path, not by the channel. (ii) When the answer is removed, the cache path loses 54% of the confidence the text path loses and its entropy rises by 73% of the text path’s rise: uncertainty crosses, attenuated—the compression of predictive variance that distillation work (Cui & Pei, 2026) predicts for any learned map. (iii) Under a contradiction the text path is swayed slightly more by the inserted sentence than by the original ($P(\text{counter}) .32 > P(\text{gold}) .29$), while the cache path keeps the original ahead (.25 > .22) and its F1 is higher (49.0 vs 45.2). Across all conditions the cache path is less confident than the text path on clean passages ($P(\text{gold}) .28$ vs .44), so a receiver that treats the two channels as equivalent would under-trust the cache; per-channel calibration is the obvious fix.

A negative result on training the loss away. Adding $\text{KL}(\text{text-path} \parallel \text{cache-path})$ on the receiver’s next-token distributions to the projector’s objective (weight 1, 1,000 steps) raises the cache path’s entropy everywhere (clean 2.38 vs 1.78) without making it track the sender better (AUROC .590 / .589 / .611; drop ratio .56 vs .54) and costs 2 F1 (Table 7, last row). Matching the receiver’s own text-path distribution is the wrong target: what is lost is the *sender’s* spread, which the receiver’s text path does not contain either. A sender-side target (matching the receiver’s cache-path entropy to the sender’s semantic entropy) is the next experiment, not this paper’s.

5. Controls

Three controls bound what the receiver actually gets from the cache, and a fourth shows why the usual headline number cannot.

Deranged cache. The *derange* arm of Table 1: a mapped cache from another episode with the same layout but a different value for the queried name. Accuracy falls to 4.6% and the receiver answers with the other episode’s value 63.7% of the time. The receiver is not ignoring the cache; it is reading it.

Deranged-sender projector. A projector trained with the identical recipe but with the sender reading text unrelated to the receiver’s continuation. It cannot learn a content channel, only a soft prompt. It reaches 23.4% on the binding test (Table 1, row 3) and 15.1 F1 on SQuAD (Table 3), both at the question-only level.

Layer-band injection. Table 2: injecting this episode’s cache into layers 0–17 alone gives chance (10.4%); layers 18–35 alone give 51.8%; no six-layer band suffices.

Continuation retention is the wrong headline. Following the closed-form-map literature (Heo et al., 2026) we also report retention (Table 8, Figure 6). A projector trained with the sender reading the *wrong* text—a learned soft prompt with no content channel—recovers 77% of the oracle gap, and almost all of it (0.72) survives when its input is deranged at test time. The trained projector’s 1.23 decomposes into ~ 0.3 of the same prompt-like effect and ~ 0.9 that disappears when the sender’s text is swapped. Retention above 1 is therefore not evidence of communication, and retention alone cannot distinguish a channel from a prompt; the binding test with its derange arm can.

6. Related Work

Cache-to-cache communication (C2C; Fu et al., 2025) and Latent Cache Flow (Rossi et al., 2026) train fusers between models of similar size; a closed-form KV map (Heo et al., 2026) reports 44–98% continuation retention across sizes; a causal audit (Cheng et al., 2026) finds that deranged caches lose only 0.4 points in such setups, which is the objection our derange and deranged-sender arms answer. A second audit (Zhang & Emu, 2026) decomposes the aggregate effect of a latent channel into “sender content” and “any message present” terms with boundary-replacement controls; our deranged-sender projector isolates the second term directly. Dual-Cache (Liu et al., 2026a), CacheBridge (Qu et al., 2026) and HyLaT (Mou et al., 2026) extend cache transfer to heterogeneous pairs and hybrid latent–text protocols. In the last six weeks, a universal context-reuse layer (Li et al., 2026) makes one model’s KV state consumable by another across scale, tokenizer and family (including

Table 7. Sender uncertainty through each handoff; 377 SQuAD items the receiver answers correctly from clean text. AUROC is of the receiver’s first-token entropy against the sender’s semantic entropy (thresholded at its median per variant); the drop ratio is the receiver’s $P(\text{gold})$ loss from clean to removed relative to the text path’s.

receiver sees	$P(\text{gold})$		entropy clean / removed	AUROC vs sender SE		drop ratio
	clean	contra. / removed		clean	contra. / removed	
text	.437	.290 / .162	1.14 / 3.03	.597	.615 / .671	1 (def.)
text + answer + confidence word (verbal)	.253	.193 / .171	2.88 / 2.80	.552	.542 / .592	.30
mapped cache	.279	.249 / .130	1.78 / 3.16	.585	.605 / .624	.54
deranged cache	.105	.106 / .105	3.88 / 3.88	.536	.488 / .564	0
zero cache	.034	.033 / .037	6.41 / 6.18	.496	.481 / .502	0
moment-matched random cache	.005	.004 / .005	7.24 / 7.30	.457	.472 / .580	0
question only	.127		3.69	.532	.479 / .603	0
mapped cache, KL-trained projector	.272	.241 / .118	2.38 / 3.76	.590	.589 / .611	.56

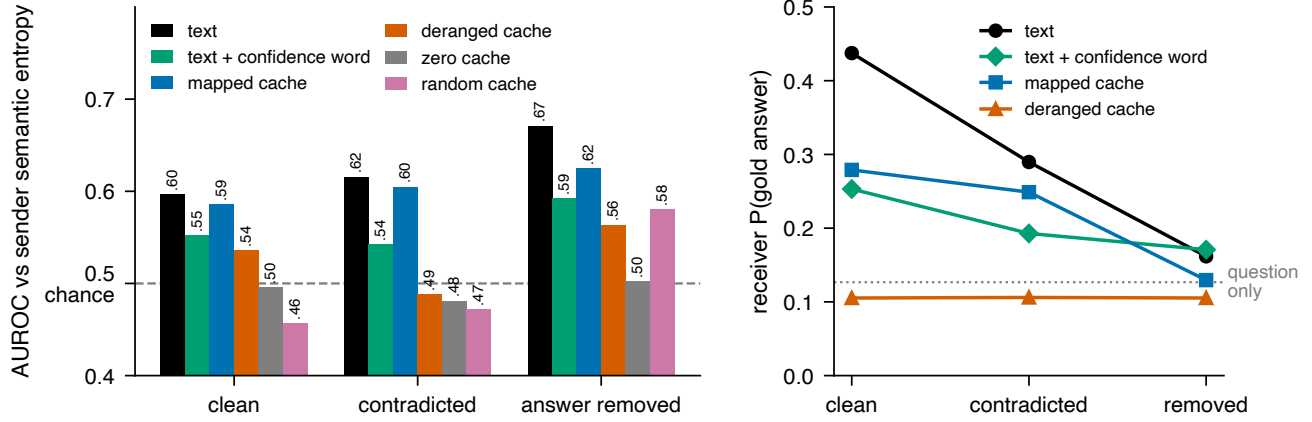


Figure 5. Left: AUROC of the receiver’s first-token entropy against the sender’s semantic entropy, per handoff and passage variant (377 items). Every content-free control sits at .46–.54; the mapped cache sits just below the receiver reading the text and above the text handoff with a confidence word. Right: the receiver’s $P(\text{gold})$ from clean to contradicted to removed passages. The cache path is less confident than the text path on clean passages and keeps the original answer ahead under a contradiction.

Table 8. Continuation retention is the wrong headline. Retention = $(\text{loss}_{\text{none}} - \text{loss}_{\text{project}}) / (\text{loss}_{\text{none}} - \text{loss}_{\text{oracle}})$ on 128 fresh held-out sequences; the second column swaps the sender’s text at test time.

projector	retention	retention, sender text deranged
ridge top-3	−0.30	−0.66
+ residual	1.23	0.31
residual, deranged sender	0.77	0.72

Llama-3.1-70B → Qwen2.5-7B) but reports no content-free control or uncertainty test; StateBridge (Peng et al., 2026) aligns final-layer hidden states into the receiver’s input-embedding space with a closed-form orthogonal map and no training, a natural training-free baseline for our fitted projector; LatentPort (Villani, 2026) hands attention KV plus recurrent state from a 4B to a 9B hybrid model without prefix replay, scored on continuation NLL rather than question answering; MentorPulse (Liu et al., 2026b) finds that a

static one-shot latent handoff can fall below no guidance on long-form generation, a scope limit our short-answer setting does not probe; and a preregistered alignment ladder between trained language-model cells (Marincat, 2026) runs a leakage-controlled causal audit close in spirit to our derange arms. The Replay Gap (Gonuguntla, 2026) and Confidence Laundering (Shi et al., 2026) argue that handoffs must be evaluated on branched rollouts and must carry uncertainty; the second is the question Section 4 answers for one channel, the first remains a next step. On the uncertainty side we borrow the knowledge-conflict construction of Longpre et al. (2021) and Xie et al. (2024), semantic entropy as the sender’s uncertainty (Farquhar et al., 2024), the finding that verbalised confidence is poorly calibrated (Tian et al., 2023; Yona et al., 2024), and the result that distillation-style maps contract predictive variance (Cui & Pei, 2026). These works are concurrent with ours; we treat them as the surrounding conversation, not as prior results this paper improves on.

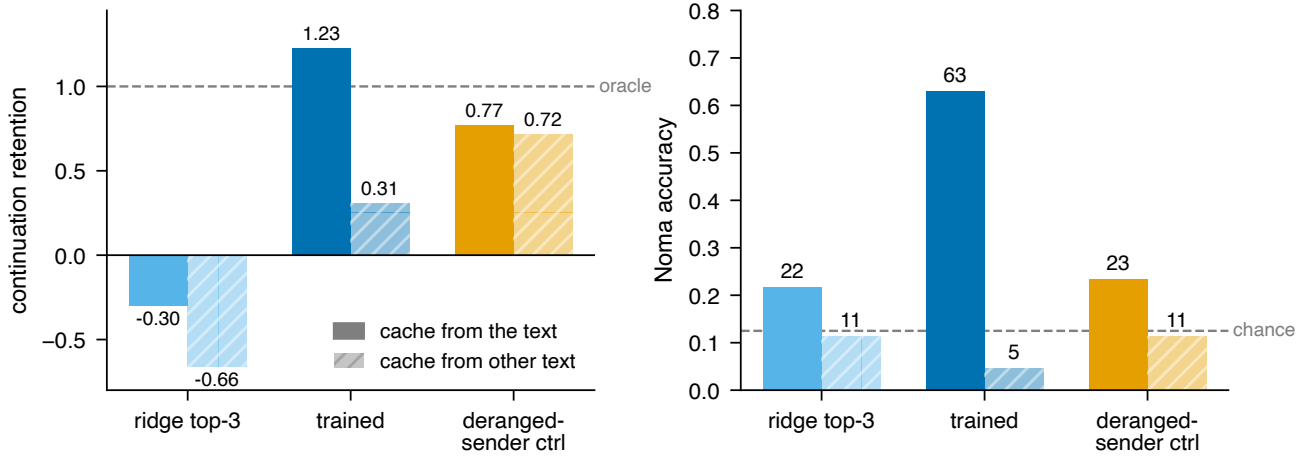


Figure 6. Retention and the binding test disagree about the deranged-sender control. Left: continuation retention on 128 fresh held-out sequences for the three projectors, with the sender reading the true prefix (solid) or a swapped one (hatched). Right: Noma accuracy for the same three projectors ($n = 1000$, 8-name variant), *project* arm (solid) and *derange* arm (hatched). Retention ranks the content-free projector at 0.77, most of which survives a swapped sender; the binding test puts it at chance.

7. Limitations

The results show that a specific fact can cross from a 3.7B cache into a 7B without text, that the crossing depends on the cache’s content, that the receiver reads it out from its later layers, and that the mapped cache keeps more of the sender’s uncertainty than a text handoff with a confidence word. They do not show that the channel is cheaper than text in bytes (144 KB per token against ~ 1 KB of text for a passage; quantisation buys $4\times$ and nothing else we tried is free), that it carries anything richer than a single fact or a short answer span, or that it survives long-form generation, where Liu et al. (2026b) report that a static latent handoff can hurt. The uncertainty result rests on 377 items of one dataset, one receiver, one sender-uncertainty measure and one design of confidence word; the absolute AUROCs are ~ 0.6 for every channel including text, so the claim is relative. The map works across families only when head dimension matches (Qwen3-4B yes, K2-Horizon-0.9B no). The first attempt to train the confidence loss away failed, and the sender-side objective that the theory box suggests is untested.

8. Conclusion

A text handoff is a point estimate and cannot carry more than $\log |Y|$ nats about how sure the sender was; a confidence word adds $\log 3$. A learned map from a small model’s KV cache to a larger model’s carries the content (63–73% binding transfer, 53.0 F1 on SQuAD, 76.7% across families) and, on top of it, more of the sender’s uncertainty than the text handoff with a confidence word does (AUROC .585–.624 vs .542–.592; drop ratio .54 vs .30). The controls matter more than the headline: the wrong cache is followed, the content-free projector fails the test, retention cannot tell a

channel from a prompt, and the signal is read out late in the receiver. What is lost is the sender’s spread, and matching the receiver’s own text-path distribution does not recover it; a sender-side target is the next experiment.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here. One consideration specific to cache handoff: a cache that a receiver “believes” (Section 5) is also a cache that can be tampered with; deployments that move caches between devices should authenticate them as they would any other input.

References

- Cheng, J., Das, S., and Ramnath, R. When does latent communication pay? a causal audit of relayed KV caches in multi-agent LLMs. *arXiv preprint arXiv:2608.04893*, 2026.
- Cui, Z. and Pei, J. How is uncertainty propagated in knowledge distillation? *arXiv preprint arXiv:2601.18909*, 2026.
- Farquhar, S., Kossen, J., Kuhn, L., and Gal, Y. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630:625–630, 2024.
- Fu, T., Min, Z., Zhang, H., Yan, J., Dai, G., Ouyang, W., and Wang, Y. Cache-to-cache: Direct semantic communication between large language models. *arXiv preprint arXiv:2510.03215*, 2025. ICLR 2026.

- Gonuguntla, A. The replay gap: Static evaluation of model switching in LLM agents scores the wrong world. *arXiv preprint arXiv:2608.08239*, 2026.
- Heo, T., Shafipour, R., Zhao, R., Golub, M., Kamani, M. M., Borkar, R., Chandran, M. T., Zardoshti, P., and Rouhani, B. D. Cross-model KV cache transfer in LLM families: A closed-form linear mapping for prefill reuse. *arXiv preprint arXiv:2608.03893*, 2026.
- Li, Y., Jiang, D., Zhao, Y., and Li, B. A universal context-reuse layer for cross-model KV sharing. *arXiv preprint arXiv:2608.30963*, 2026.
- Liu, J., Zhang, Q., Jia, Y., Kan, Z., and Wang, S. Dual-cache latent space communication between heterogeneous language models. *arXiv preprint arXiv:2608.20617*, 2026a.
- Liu, Z., Li, G., Qiu, C., Kong, W., and Kalnis, P. MentorPulse: Refreshing cross-model latent guidance for long-form generation. *arXiv preprint arXiv:2608.20927*, 2026b.
- Longpre, S., Perisetla, K., Chen, A., Ramesh, N., DuBois, C., and Singh, S. Entity-based knowledge conflicts in question answering. In *Proceedings of EMNLP*, 2021. arXiv:2109.05052.
- Marincat, N. Portable semantics, private dialects: Reuse and negative transfer in latent communication between language-model cells. *arXiv preprint arXiv:2609.11365*, 2026.
- Mou, X., Wang, S., Li, Z., He, Y., and Wei, Z. HyLaT: Efficient multi-agent communication via hybrid latent-text protocol. *arXiv preprint arXiv:2605.25421*, 2026.
- Peng, Y., Zhang, D. C., Wang, X., and Aletras, N. State-Bridge: Training-free hidden-state alignment for latent communication in LLM multi-agent systems. *arXiv preprint arXiv:2608.13317*, 2026. COLM 2026.
- Qu, X., Lu, S., Chen, Z., Wang, S., and Lin, T. CacheBridge: Efficient cross-model KV cache transfer. *arXiv preprint arXiv:2609.00891*, 2026.
- Rossi, M., Raghunath, P., and Wu, E. Latent cache flow: Model-to-model communication without text. *arXiv preprint arXiv:2605.22863*, 2026.
- Shi, K., Zhang, Z., Bao, H., Nelson, C., and Ye, Y. Confidence laundering in agent systems: Why uncertainty needs a latent carrier. *arXiv preprint arXiv:2606.20662*, 2026.
- Tian, K., Mitchell, E., Zhou, A., Sharma, A., Rafailov, R., Yao, H., Finn, C., and Manning, C. D. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of EMNLP*, 2023. arXiv:2305.14975.
- Villani, S. P. LatentPort: Beyond KV cache – cross-model transfer of recurrent memory in hybrid language models. *arXiv preprint arXiv:2609.25053*, 2026.
- Xie, J., Zhang, K., Chen, J., Lou, R., and Su, Y. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts. In *Proceedings of ICLR*, 2024. arXiv:2305.13300.
- Yona, G., Aharoni, R., and Geva, M. Can large language models faithfully express their intrinsic uncertainty in words? In *Proceedings of EMNLP*, 2024. arXiv:2405.16908.
- Zhang, H. and Emu, M. Do latent channels actually communicate? a causal audit of latent multi-agent LLM. *arXiv preprint arXiv:2607.26773*, 2026.

A. Ablations on the 6-Name Variant

All runs: sender K2-Horizon-3.7B, receiver K2-Horizon-7B, colour attribute, 6 names, $n = 300$ episodes, chance 12.5%. Accuracy in percent.

Table 9. Projector ablations on the 6-name Noma variant.

projector	project	derange	follow
ridge top-3 (closed form)	22.0	10.7	30.3
+ residual, 300 steps	56.3	6.3	53.7
+ residual, 1500 steps, seed 0	71.3	4.3	71.7
+ residual, 1500 steps, seed 1	73.0	3.0	74.0
+ residual, 1500 steps, seed 2	75.3	2.3	76.0
+ residual, 1500 steps, ridge frozen	59.0	7.0	59.3
+ residual, 1500 steps, aligned-layer ridge	32.7	8.0	34.7
+ residual, 1000 steps, bottleneck 32	64.3	5.7	62.3
+ residual, 1000 steps, bottleneck 16	61.3	7.7	60.0
+ residual, 1000 steps, + KL(text cache)	66.0	5.3	63.3

B. Other Attributes and Reversed Roles

Table 10. Binding transfer with city and animal attributes ($n = 500$, 8 names) and with the roles reversed (7B sender, 3.7B receiver; 6 names, $n = 300$).

setting	projector	none	project	derange	follow
city	ridge top-3	14.2	19.8	11.8	18.4
city	trained	14.2	61.8	6.4	60.2
city	deranged-sender ctrl	14.2	15.4	11.4	18.6
animal	ridge top-3	12.2	12.8	12.0	11.2
animal	trained	12.2	67.2	5.2	64.6
animal	deranged-sender ctrl	12.2	12.4	12.8	9.8
reversed	ridge top-3	11.7	29.0	12.3	31.3
reversed	trained	11.7	63.3	5.7	63.0

C. Distance Sweep and Layer Bands

Table 11. Noma accuracy (*project* arm) versus filler tokens between facts and question. 8-name variant; $n = 1000$ at pad 0, $n = 500$ otherwise.

projector	0	64	128	256	512
ridge top-3	21.7	22.2	22.4	22.8	24.2
trained	63.0	61.0	57.0	62.4	68.8
deranged-sender ctrl	23.4	25.0	19.8	18.2	13.8

D. Compression Sweep

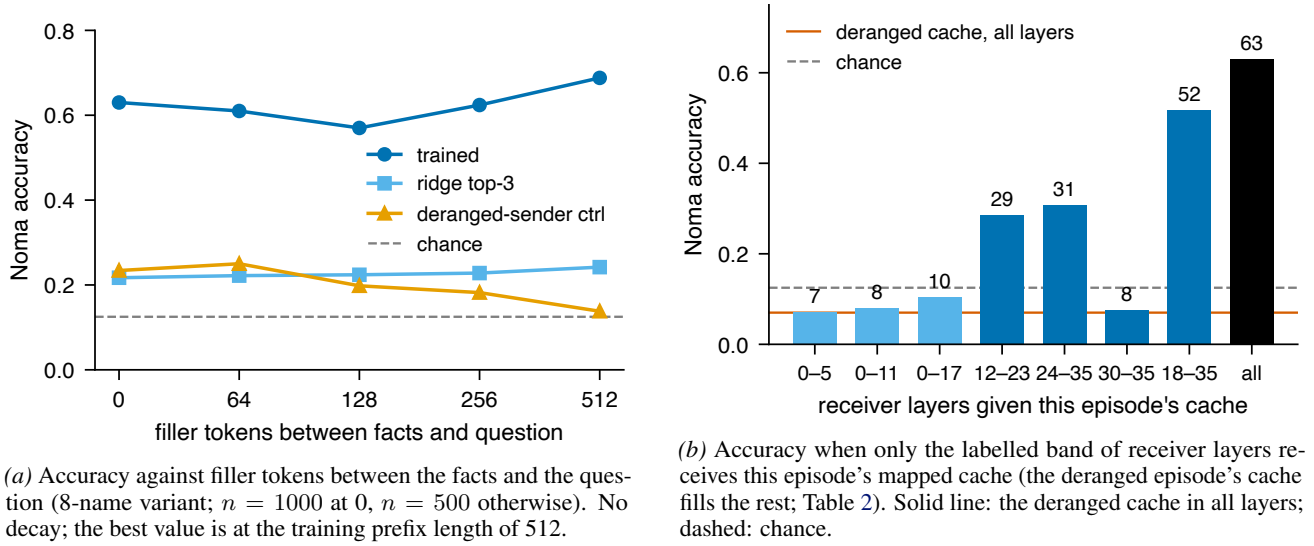


Figure 7. Distance sweep and layer-band injection.

Table 12. Every compression of the seed-2 projector's mapped cache. KB per token is for 512-token SQuAD passages; low-rank truncation and mean-vector fill have a per-sequence overhead, so their per-token size depends on sequence length (for the 29-token Noma facts, rank 32 exceeds the full cache). Noma: 6-name variant, $n = 300$, percent; SQuAD: $n = 300$, token-F1.

spec	KB / token	Noma project	Noma derange	SQuAD F1
none (full, bf16)	144	75.3	2.3	53.0
int8	72	76.0	2.7	52.1
int4	36	75.0	3.7	51.6
rank 32	78.7	75.0	2.3	45.3
rank 16	39.3	75.0	2.3	29.6
rank 8	19.7	51.7	7.0	17.2
layers 18-35, zeros	72	24.3	9.7	8.3
layers 18-35, zeros, int8	36	24.7	9.7	7.8
4 heads, zeros	72	25.7	8.7	8.2
2 heads, zeros	36	22.3	9.7	5.9
layers 12-35, mean fill	96.4	35.0	11.0	28.8
layers 18-35, mean fill	72.7	33.7	11.7	21.4
layers 18-35, mean fill, int4	18.7	33.3	11.7	21.1
layers 24-35, mean fill	48.9	29.3	12.0	14.5
4 heads, mean fill	72.7	23.0	11.3	15.3